INCH-POUND

MIL-P-70533 (AR)
29 OCTOBER 1990

MILITARY SPECIFICATION

PROJECTILE 105MM, HE, RA XM913/XM927
ROCKET PROPELLANT FOR

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for the fabrication and packaging of the rocket propellant used in the XM913 and XM927, 105MM, HE, RA, Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specifications for
MIL-C-70532	- Cartridge, 105MM, HE, RA, XM913 Loading, Assembling and Packing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315

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STANDARDS

MILITARY

MIL-STD-286	- Propellants Solid Sampling, Examination and Testing
MIL-STD-410	- Non-destructive Testing Personnel Qualification and Certification
MIL-STD-453	- Inspection, Radiographic

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from Military Specificaiton and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKING DRAWINGS

9381132	- Grain, Propellant
9381151	- Grain, Spacer

(Copies of drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Code of Federal Regulations

Title 49	- Transportation parts 100-199

(Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)".)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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CHEMICAL PROPULSION INFORMATION AGENCY

CPIA Publication No. 21 - JANNAF Solid Propellant Mechanical Behavior Manual (Section 4.3.2, Uniaxial Tensile Test of Constant Strain Rate; Section 4.7.5 Propellant Liner Peel Test, Nov. 70)

(Copies of the above document(s) required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Propellant grain. The inhibited propellant grain shall comply with the requirements specified on 9381132 and with all requirements specified in applicable specifications.

3.3 Process controls. The contractor shall identify the Critical processes and determine the necessary control parameters to assure an acceptable product. Process controls shall be submitted to the Government for approval (see 6.8).

3.3.1 Mandatory controls. During mixing operations the mixing bays shall be maintained at a temperature of 70°F to 80°F with a maximum relative humidity of 50%.

3.4 Heat of reaction. The heat of reaction of each batch of propellant shall be 1550 ± 70 calories per gram.

3.5 Inhibitor peel strength. The minimum peel strength of the inhibitor/propellant bond at 70° ± 5°F shall be 20 pounds per linear inch.

3.6 Strand burning rate. Strand burning rate shall be in accordance with Table I.

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TABLE I. Strand Burning Rate

Temperature (°F)	Pressure (PSI)	Burning Rate (inches/seconds)
70	1000	0.37 + 0.05
70	2000	0.47 + 0.05
145	2000	0.50 + 0.05
-50	1000	0.31 + 0.05

3.7 Physical defects. Each grain segment (individual split) shall be subjected to visual and radiographic inspection. Segments with defects in excess of those listed in 3.7.1 or 3.7.2 shall be rejected (see 6.5).

3.7.1 Radiographic inspection.

a. Cracks. There shall be no cracks in excess of 0.06 inches in length as evidenced by dark lines on the radiograph.

b. Voids. There shall be no voids measuring more than 0.18 inches in any direction. The total combined length of all voids shall not exceed 0.25 inches. All voids, regardless of size, shall be counted towards the combined total length. Voids will appear as low density or rounded shadows on the radiograph.

c. Foreign material. There shall be no foreign material measuring more than 0.06 inches in any shape or dimension as evidenced by bright spots on the radiograph.

3.7.2 Visual examination.

a. Cracks. There shall be no cracks in excess of 0.06 inches in length.

b. Voids. There shall be no voids or bubbles measuring more than 0.18 inches in any direction. The total combined length of all voids shall not exceed 0.25 inches.

c. Foreign material. There shall be no foreign material in excess of 0.09 inches in any shape or dimension.

3.7.3 Inhibitor separation. On finished propellant segments there shall be no visual evidence of inhibitor/propellant separation (delamination) within one inch of the edge of any inhibitor/propellant mating surface. In addition there shall be no visual evidence of bubbles or separations in the inhibitor/propellant bond that are, individually, greater than 0.01 square inch in area or collectively greater than 0.02 square inch in total combined area.

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3.7.3.1 Inhibitor flash. There shall be no evidence of inhibitor flash as indicated on the applicable drawing.

3.8 Mechanical properties. The cured propellant shall have the mechanical properties specified in Table II.

TABLE II. Mechanical Properties of Propellant

Characteristics	70° F (± 5° F)	-50° F (± 5° F)	145° F (± 5° F)
Modulus of elastricity (PSI)	2,485 ± 15%	13,170 min	1,636 min
Ultimate Tensile Stress (PSI)	204 ± 10%	455 min	139 min
Strain at Max. Stress (%)	28.2 ± 15%	12.4 min	13.7 min
Ultimate Strain (%)	29.3 ± 15%	16.5 min	14.4 min

3.9 First article inspection. This specification contains technical provisions for the first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.9.1 Proving ground tests (first article only). When tested in accordance with MIL-C-70532 the rocket grains shall meet the requirements of MIL-C-70532.

3.10 Proving ground tests. When assembled to projectiles the grain shall ignite, burn and enable the projectile to achieve its target range. The mean horizontal range, after all applicable corrections specified in the current available firing tables for the XM913, shall not be less than 98 percent of the range specified in those firing tables for the weapon elevation fired.

3.11 Workmanship. Prior to casting, the batch of propellant shall be homogeneous and there shall be no streaks or discoloration signifying improper or incomplete blending of ingredients. The finished propellant grain shall be free of cracks, dirt, grease, gouges, foreign matter or any other physical defects which would render the item unsuitable for its intended use. Marking shall be complete and legible.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection.
b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.1.2. The first article sample shall consist of the following items in sample quantities as indicated:

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Description	Drawing	Quantity
Spacers	9381151	10
Strand burning specimens		12
Bulk sample of sufficient size for 5 tensile and 2 heat of reaction specimens	9381132	1
Inhibitor peel strength specimens	9381132	2
Propellant grain (finished)	9381132	30 segments

The first article sample shall be manufactured using the same approved process control (see 3.3), equipment and procedures as will be used in subsequent production and, all materials shall be obtained from the same source of supply as will be used in regular production. Test equipment and standards, as required, shall be approved by the acquiring activity prior to submission of the first article sample.

4.3.2 Inspections to be performed. See MIL-A-48078 and the first article inspection table herein. Unless otherwise directed by the Contracting Officer, the propellant grains used for ballistic testing shall not be submitted prior to acceptance of the "Inspection and tests (non ballistic)" samples submitted to the Government.

4.3.2.1 Tests to be performed. Proving ground tests shall be performed using the procedures of 4.5.7.

4.3.3 Rejection. See MIL-A-48078.

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Table III. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9381132
	XM913 Propellant Components		NEXT HIGHER ASSEMBLY 9381147

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Inspection and Tests (non ballistic)			
	Spacer			
	(Dwg. 9381151)	10 each	3.1	4.4.2.1
	Propellant After Mixing and Curing			
	(Dwg. 9381132)			
	Mechanical properties	5 each	3.8	4.5.2
	Heat of reaction	5 each	3.4	4.5.3
	Inhibitor peel strength samples	2 each	3.5	4.5.4
	Strand burning rate (strands)	12 strands	3.6	4.5.1
	Complete Rocket Grain			
	(Dwg. 9381132)			
	Radiographic inspection	10 each	3.7.1	4.5.5/4.4.2.4
	Examination for defects	10 each	3.2	4.5.5/4.4.2.4
	Testing			
	Ballistic test	10 grains/ lot	3.10	4.5.7

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation.

a. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

b. In addition, each lot of propellant grains shall consist of propellant and inhibitor produced from single lots of basic materials by the same process with approved process controls in not more than two consecutive months. Individual propellant batches, once accepted, shall be combined into a grain lot.

4.4.1.1 Propellant batch. A batch of material, both inhibitor and propellant, is defined as a homogeneous quantity of raw materials that are blended and processed at one time. As a maximum, a batch shall consist of materials blended at one time in one mixing bowl. A batch shall contain individual ingredients from not more than one lot from one manufacturer for each ingredient.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Attributes sampling inspection shall be conducted in accordance with Table IV below, using the inspection levels cited in the Classifications of Characteristics paragraphs.

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TABLE IV. 0 - 1 Attributes Plans

Lot Size	Inspection Levels I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
8 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10

Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative inspection provisions. Alternative inspection procedures, methods or quipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70533 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Spacer, Grain	1 OF 1	9381151
			NEXT HIGHER ASSEMBLY: 9381132

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness	Level III	3.1	Gage
102	Diameter	Level III	3.1	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.11	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Propellant After Mixing and Curing	1 OF 1	9381132
			NEXT HIGHER ASSEMBLY 9381147

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Mechanical properties	5 100%	3.8	4.5.2
102	Strand burning	12 100%	3.6	4.5.1
103	Inhibitor peel strength	1 100%	3.5	4.5.4
104	Heat of reaction	2 100%	3.4	4.5.3
Minor	None defined			

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70533 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Grain After Assembly	1 OF 1	9381132
			NEXT HIGHER ASSEMBLY 9381147

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special 1	Radiographic inspection*	100%	3.7.1	4.5.5
Major 101	Radius of corner	Level III	3.2	Gage
102	Total height across three segments	Level III	3.2	Gage
103	Spacer missing (6 places)	Level III	3.2	Visual
104	Length of taper	Level III	3.2	Gage
105	Diameter of outside of grain	Level III	3.2	Gage
106	True position of space between grain segments	3.12	3.2	Gage
107	Inhibitor perforated, split or separated	100%	3.7.3	Visual
108	Cracks, porosity or foreign material	100%	3.7.2	Visual
Minor 201	Evidence of poor workmanship	Level V	3.11	Visual

NOTES: * At the option of the contractor the grain may be radiographically inspected as individual splits in lieu of as an assembly.

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4.4.3 Testing. Testing is described in the Quality Conformance and first article tables and herein.

4.4.3.1 Proving ground. These tests shall be performed as specified in 4.5.6. Defects shall be classified as follows:

Critical	None defined
Major	
101	Range, minimum
Minor	None defined

4.4.4 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein, the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See MIL-A-48078 and 6.4 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification or calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by 4.4.4.

4.4.4.2 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with MIL-STD-410.

4.5 Test methods and procedures.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Strand burning characteristics of propellant. Along with propellant grains cast from each batch or mix of chemicals, 12 test strands shall be taken from the same conditions for the same length of time and subjected to this test in accordance with Method T803.1 of MIL-STD-286. Three strands will be tested at each of the temperature, pressure conditions in Table I of paragraph 3.6.

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4.5.2 Mechanical properties of propellant. Along with propellant grain cast from each batch or mix of propellant, one physical property sample, sufficiently sized shall be cast from the same propellant batch, cured in the same curing chamber under the same conditions for the same length of time and subjected to these tests in accordance with the JANNAF Class B procedures as described in CPIA Publication #21 "Uniaxial Tensile Tests" at constant crosshead rate. Test specimens shall be conditioned at the applicable test temperature for a minimum of two hours prior to performing the mechanical property tests. The crosshead speed of the tensile tester shall be 2.0 inches per minute and the chart speed shall be 10 inches per minute. Test specimens shall be cut from a bulk sample. Five test specimens shall be prepared and tested at each test temperature. If the average value of the five test specimens fail to meet the requirements of 3.8, the batch of propellant of which it is representative shall be rejected.

4.5.3 Heat of reaction. One representative propellant specimen obtained from the same physical property sample prepared for the mechanical property tests shall be used to perform this test accordance with Method 802.1 of MIL-STD-286 or an approved equivalent. The 3 grain sample and oxygen purge requirements shall not apply.

4.5.4 Inhibitor peel strength. Two (2) grain segments shall be randomly selected to represent each batch of propellant for this test. One specimen shall be prepared from each segment and tested in accordance with 4.7.5 of CPIA Publication 21. If either specimen fails to meet the requirement of 3.5, the grains prepared from the batch of propellant represented shall be rejected. The sample segments may be selected from pieces rejected for causes not related to the inhibitor.

4.5.4.1 Reduced sampling. After successful acceptance of ten propellant batches, the inhibitor peel strength sampling shall be reduced to one grain segment from each propellant batch. Failure of any specimen shall require rejection of grains from that propellant batch. Sampling will revert to the initial sampling sequence of 4.5.4 until sampling can again comply with the above reduction criteria.

4.5.5 Tool control dimensions. Each tool (mold) shall be gaged before its first usage in production and before every tenth usage thereafter. The tools shall be gaged for first article and whenever a tooling change, adjustment or repair is made. The first component from each tool shall be measured and the results

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recorded for informational purposes. Each tool shall be visually examined for foreign material or damage before each usage. If foreign matter is present, it shall be removed. If damage is suspected, the tool shall be gaged and removed if out of tolerance.

4.5.6 Physical defects. The finished propellant grain shall be radiographically inspected for physical defects in accordance with MIL-STD-453 with the beam perpendicular to the longitudinal axis. A penetrameter or sample test block showing the types of defects for rejection shall be designed by the contractor to assure a two (2) percent minimum sensitivity and approved by the acquiring activity (see 6.5). Finished propellant grain assemblies which exhibit any of the physical defects specified in 3.7 shall be rejected and removed from the lot.

4.5.7 Proving ground. This test shall be conducted in accordance with an applicable proving ground test procedure. All projectiles fired for this test will be assembled with a point detonating fuze.

The projectiles fired in this test shall be fired in the M119 Howitzer using a XM229 series propelling charge. The projectile, fuze and propelling charge shall all be at ambient temperature. A constant gun QE will be maintained throughout the test. Smear photographs will be taken at 30 meters in front of the muzzle. Doppler radar (HAWK) will be used.

An effort will be made to fire all projectiles for this test within a one hour time span. Observation shall be made to determine compliance with 3.10. Stop the test if surface winds exceed 20 knots or if two consecutive wind speed readings differ more than 15 knots or if the wind direction of two consecutive readings differs by more than 800 mils (does not apply if wind is less than 5 knots).

4.5.7.1 Weapon and component data. All weapon and ammunition components used in support of this test shall be identified by model and lot or serial numbers.

4.5.7.2 Data reporting. As a minimum, the proving ground test report will include the following observable attribute data:

a. Premature functioning
b. Metal parts separation
c. Impact functioning (information only)
d. Round by round range
e. Mean range

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5. PACKAGING

5.1 Packaging. Packaging and packing shall be in accordance with Code of Federal Regulations (CFR), Title 49, Parts 100-199.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of inspection procedures and equipment design for approval. See MIL-A-48078. Submit equipment designs as required to: Commander, ARDEC, ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Lot sizes. It is noted that the size of inspeciton lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.6 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.3	Process control	DI-P-1604
4.4.4	Inspection equipment approval	DI-R-1714
4.5.2	Test report	DI-T-1906

(Copies of data item descriptions required by the contractor in conjunction with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.7 Objective evidence.

6.7.1 General. Inspection records and documentation verifying compliance with applicable drawing and specification requirements.

6.7.2 Materials. A statement (certification), supported by test data, that materials produced or purchased by the contractor meets all requirements when such material is controlled by Government or commercial specifications referenced in contractual documents.

6.8 Process controls. Process controls and process control changes should be submitted for approval to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-FSA-IM, Picatinny Arsenal, New Jersey 07806-5000. Process controls shall specify in detail all materials, procedures, techniques and equipment (including handling of ingredients, mix cycle, sequences of addition of materials, mixing times and conditions, cure temperatures and times, etc.) required to process, fabricate, inspect and test a quality product suitable for its intended use as determined by the acquiring agency. Detailed test and inspection record format shall be included as an integral part of the process and shall be in sufficient detail as to enable easy verification by the Government QAR designated by the acquiring agency.

6.9 Physical defects. Individual segments failing the requirements of 3.7.1, 3.7.2 or 3.7.3 shall be rejected and removed from the lot. The remaining acceptable segments, from one batch of propellant, may be matched with other acceptable segments to form grain sets.

6.10 Subject term (keyword) listing.

Inhibitor
Process controls
Propellant grain
Radiographic inspection
Reduced sampling
Strand burning

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1315-AC29)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70533 (AR)	29 October 1990

3. DOCUMENT TITLE

PROJECTILE 105MM, HE, RA XM913/XM927 ROCKET PROPELLANT FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
US Army ARDEC Standardization & Specification Office	(201) 724-6671	880-6671
ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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